



Synopsys' Galaxy implementation platform

information within the Milkyway database for parasitic extraction required for timing analysis and verification.

IC Validator In-Design is foundry-qualified and can also be used with both IC Compiler and Galaxy Custom Designer for physical verification of cell-based and custom circuits. Numerous 20nm tapeouts have already been completed using IC Validator for physical verification. The in-design capabilities along with IC Compiler help to shorten design closure time.

Kalra shares, "Increasing the number of masks and processing steps increases the marginality and variability in addition to increasing the manufacturing cost and time. It needs to be managed at the design level not only around transistors but at metal levels, requiring designers to take care of the local layout effects upfront during schematic designs with effective placement and routing. These device and metal variabilities need to be modeled in PDK, libraries and flow to have an effective signoff. With next-generation SoCs becoming multi-core with multiple interfaces and sensors, designers also need to understand the various sources of variability to exploit the technology options to its limits with appropriate EDA tools."

3 Achieving higher levels of extraction and timing analysis

20 nm means accurate higher levels of extraction and timing analysis to allow manufacturing variability.

Ahuja informs that one new source

of manufacturing variability at 20 nm is double patterning. Different masks on a given layer will often shift during the manufacturing process, resulting in variations that impact capacitance. Extraction software must take mask offset into account when extracting RC values.

Another source of variability is layout-dependent effects. At 20 nm, the layout context (what is placed near a device) can impact device performance by as much as 30 per cent. Thus circuit designers need to consider layout context as well as device topology, and simulate with layout effects prior to layout completion. This requires pre-layout sensitivity analysis tools, as well as context-driven placement and optimisation.

Biddle adds that as some of the critical routing layers in 20nm designs will be constructed out of two separate masks, there is a concern that shifts in alignment due to tolerances in the manufacturing process could lead to parasitic variations with resultant timing variations. One way to solve this variation is to add additional process corners to analyse. However, this will have a potentially huge impact on design turnaround time and is therefore not a preferred solution.

"While the full impact of this will not be known until the industry has gained more experience and the foundries optimise their manufacturing techniques, we have built a foundry-recommended capability into our extraction and static timing analysis tools StarRC and PrimeTime in case it is required," he added.

The Standard Parasitic Exchange Format (SPEF) file produced by extraction tools has been extended to support extraction of multiple parasitic capacitances. The new M-SPEF format provides the capability to capture minimum and maximum values, in addition to the extracted nominal para-

sitic capacitance extracted at each PVT corner. StarRC has been enhanced to support the multi-value SPEF and also the new format, and has been certified for 20 nm. The information for modeling this variability effect is provided in source technology files with StarRC producing these.

4 Capacity, performance requirements

There are certain performance and capacity requirements for next-generation designs.

According to Ahuja, 20nm ICs may have 8-12 billion transistors. Advanced node technology thus requires performance, power and capacity breakthroughs that should allow electronics OEMs to provide next-generation designs. To meet time-to-market and PPA (performance, power and area) goals for extremely large and fast chips, design productivity must improve multi-fold.

Faster tools are part of the solution. For example, GigaOpt technology in Encounter toolset provides optimisation across the IC physical design flow, runs on multi-core computers and delivers a 30 per cent run-time speedup on a single CPU.

The other part of the solution is to move up in abstraction level. This is where Encounter GigaFlex technology comes in. It allows designers to model only what they need at high abstraction levels, and add details as the design proceeds and more accuracy is required.

Biddle shares, "We are already seeing Synopsys' Galaxy Implementation Platform being deployed on designs with over a hundred million gates in addition to billions of transistors for memory. We also see chips designed to perform at greater than 2 GHz. As these chip sizes and performances, leakage power and active power management continue to be a challenge as designers try to harness the performance capabilities of 20 nm while still meeting their power budgets." ●

The author was executive editor at EFW till recently